GORENSTEINNESS OF A LOCAL RING IS PRESERVED AND REFLECTED UNDER FINITELY GENERATED FIELD EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0C03 (sources/stacks-duality/pinned-original-source.tex line 6480).

Allowed standard prerequisites (editorial).

- Descent and ascent of Gorensteinness along a faithfully flat local homomorphism. Dualizing Complexes, Lemma labelled dualizing-lemma-flat-under-gorenstein, cited at proof lines 6518-6524. Gives both directions once the fibre is known to be Gorenstein.
- Formal fibres of Gorenstein ring maps. Dualizing Complexes, Lemma labelled dualizing-lemma-formal-fibres-gorenstein, cited at proof lines 6530-6532. Shows that the tensor product fibre is Gorenstein.

Source reference. Original lemma 0BJL: sources/stacks-chow/tags.txt, source lines 14529–14529. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6499 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BJN: sources/stacks-chow/tags.txt, source lines 14531–14531. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6511 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a locally Noetherian scheme over the field k. Let k'/k be a finitely generated field extension. Let $x \in X$ be a point, and let $x' \in X_{k'}$ be a point lying over x. Then we have*

$$\mathcal{O}_{X,x} \text{ is Gorenstein} \Leftrightarrow \mathcal{O}_{X_{k'},x'} \text{ is Gorenstein}$$

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If X is locally of finite type over k, the same holds for any field extension k'/k.

Proof. In both cases the ring map $\mathcal{O}_{X,x} \to \mathcal{O}_{X_{k'},x'}$ is a faithfully flat local homomorphism of Noetherian local rings. Thus if $\mathcal{O}_{X_{k'},x'}$ is Gorenstein, then so is $\mathcal{O}_{X,x}$ by Dualizing Complexes, Lemma 0BJL. To go up, we use Dualizing Complexes, Lemma 0BJL as well. Thus we have to show that

$$\mathcal{O}_{X_{k'},x'}/\mathfrak{m}_x\mathcal{O}_{X_{k'},x'} = \kappa(x) \otimes_k k'$$

is Gorenstein. Note that in the first case $k \to k'$ is finitely generated and in the second case $k \to \kappa(x)$ is finitely generated. Hence this follows as property (A) holds for Gorenstein, see Dualizing Complexes, Lemma 0BJN. □